Strong internal results; uneven external performance

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Question and approach

Can a Cleveland-trained model classify existing coronary artery disease across historical hospital cohorts? This educational study compared logistic regression, Random Forest, and XGBoost with fold-fitted preprocessing. Development-only calibration and cutoff selection froze a 200-tree Random Forest (c06), with no added sigmoid calibrator, at probability >=0.3528716233973672. Twelve later-workup predictors exclude ca; record-level timing before diagnosis remains unverified.

Final results: same model and cutoff at every site

Cohort	n	ROC-AUC (95% interval)	Sensitivity	Specificity
Cleveland test	61	0.957 (0.907-0.990)	92.9%	78.8%
Hungarian	293	0.903 (0.864-0.939)	92.5%	67.4%
Switzerland	123	0.766 (0.591-0.925)	88.7%	50.0%
VA Long Beach	199	0.721 (0.639-0.799)	93.9%	23.5%

The internal test missed 2/28 positive records and flagged 7/33 negatives. VA detected 139/148 positives but flagged 39/51 negatives. Swiss sex=0 sensitivity was 3/10 (95% interval 10.8%-60.3%); specificity was undefined with no negatives. These are descriptive findings, not a fairness certification.

What the results mean

External transportability is the central finding. Strong internal discrimination did not translate into consistent specificity or probability calibration across sites. Swiss cholesterol was entirely missing; other external workup inputs were frequently unavailable. These shifts may matter, but no causal explanation was established. SHAP highlighted thal, cp and slope as prediction contributors, not biological causes. No clinical benefit or cost saving was measured.

The 90% sensitivity objective was exploratory. The nested development selection policy achieved 88.3%; final internal sensitivity's interval was 77.4%-98.0%. Neither the final point estimate nor selection scores guarantee a population target. Intervals condition on the fitted model and exclude selection uncertainty.

Delivery and remaining verification

The report, model card, saved predictions and evidence verifier are packaged. The frozen model and cutoff remain unchanged after test exposure. Historical referral data, sparse groups and uncertain feature timing limit intended use to education and technical review.

Accepted Windows binary/full environment exports remain unavailable to independent review; clean-environment artifact reproduction and tagged release remain pending. External raw/demographic memberships were not independently reloaded in the numerical review. The site-comparison PNG hash discrepancy remains unresolved.

Evidence: reports/day09/report.pdf; reports/day08/site_metrics.csv; reports/day08/review_checks.json; docs/model_card.md; docs/reproducibility.md. Dataset: Janosi et al., UCI Heart Disease, DOI 10.24432/C52P4X.